

Gas Absorption in a Packed Bed

Final Lab Report

CHE382: Unit Operations Lab

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Group **3**

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Executive Summary

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In the chemical industry, one-unit operation that is commonly used is gas absorption. It enables the removal of a particle from a gas stream, typically done so in a packed column. The packing of columns varies with applications, such as activated carbon for adsorption or Raschig rings for absorption. Mass transfer is achieved in gas absorbers when a soluble vapor comes into close contact with a liquid within the column. The process is driven by the differential partial pressure of the gas in the mixture and the vapor pressure above the liquid's surface.

In this experiment, the objective was to determine the overall mass transfer coefficient in the removal of CO₂ from air varying water and air flow rates. The fluid flow characteristics in a packed bed was also determined along with the flooding point.

Four different water flow rates of 1.5, 3.0, and 6.0 L/min were used to determine the flooding point with air entering the system at varying flows (see Table 3, Appendix I). Flood points were found to decrease with increasing water flow. The observed flood point for 1.5 L/min of water was at an air flow of 120 L/min and decreased to 30 L/min at a water flow of 6 L/min. Even though the experiment called for water at 8 L/min, this experiment was not done as the column started flooding uncontrollably. As the experiment progressed, CO₂ was introduced into the system at 3.0 L/min. From the concentrations of CO₂, low values of the mass transfer coefficients, k_a , were calculated. This shows a low transfer of CO₂ into water and air.

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Introduction

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Absorption columns are widely used in the chemical industry and can play an integral part in a chemical engineer's career. For this reason, it is important for engineers to understand the behaviors observed in such process units to be able to properly design and operate them. Gas absorption is defined to be a unit operation where one or more components of a gas stream are removed through absorption into a nonvolatile liquid, also known as the solvent. The liquid solvent is known as the separating agent and the adsorption itself can be performed through physical or chemical routes.[1] The physical absorption process relies on solubility levels of the gaseous components; the more soluble a component is the more of it will be absorbed by the liquid solvent. Examples of physical absorption in the chemical industry include removal of butane and pentane from refinery gas and oil mixtures.[1]

On the other hand, the chemical absorption process involves the removal of components in the gas phase through reactions that occur with the solvent and allow the gaseous components to remain in solution.[1] This particular process has become very important to the chemical industry because it is used for the removal of CO_2 and H_2S gases from process stream. Absorption processes like this are relevant in today's society due to the rise in climate change concerns as a result of CO_2 buildup in the atmosphere.

This experiment utilizes the Armfield Gas Absorption Column which is packed with 1-cm Raschig rings, where that inner diameter of the column is 8 cm and the outer and inner diameters of the Raschig rings are 1.0 cm and 0.8 cm respectively. The absorption unit is equipped with liquid flow system that includes a centrifugal pump and a rotameter that read the flowrate of the liquid and gas streams. Along with that, a Hempl Apparatus is attached to the unit which allows for accurate gas analysis.

The first objective of this experiment is to determine the fluid flow characteristic that can be observed in a packed bed gas absorber; including the behavior of the flooding point. This is carried out in the first portion of the experiment where the void volume of the packed bed is determined and then pressure drops across the absorption column are measured at various air flowrates using manometers. The air flowrates are adjusted until the column reaches its flooding point.[2]

The second objective of this experiment is to determine the behavior of gas absorption and calculation of the overall mass transfer coefficient when CO₂ gas is removed from a stream of air with water being used as the liquid solvent. [2] This is done in the second and third portion of the experiment where carbon dioxide is introduced into the air stream. The second portion focuses on bringing the water stream to an equilibrium with the carbon dioxide in the air stream and then performing an analysis on the dissolved gas in the liquid. Then the carbon dioxide gas is desorbed back into the lab air. The third portion of the lab focuses on the absorption of the carbon dioxide into the water solvent while the column is operated in a continuous mode. In both the second and third section, liquid samples are taken from the column and analyzed through titration to calculate the composition. Gas samples are also taken and analyzed using the Hempel Apparatus to determine the composition of the gas. [2]

Theory

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Fluid Characteristic in a Packed Bed Gas Absorber

Understanding the fluid characteristics is important to chemical engineers because it allows for proper design of the absorption column. For example, being able to accurately describe a pressure drop trends in the column will allow for accurate sizing and design of pumps. In the experiment the pressure drop across the packed bed is measured as a function of the air and water flow rates. [2]

Furthermore, understanding the correlation between gas and liquid flowrates, and column packing material is crucial to avoid flooding of the absorption column. Flooding occurs when the gas flow rate is much greater than the liquid flowrate. This means the gas can push through the column and lift out of the packed bed decreasing the contact between the gas and liquid phase and decreases the effectiveness of the absorption.[2]

When the packing inside the absorption column is dry and the liquid flow rate is zero ($L=0$), the pressure drop increases as the velocity of gas increases and a linear relationship can be observed when on a log-log plot. This relationship is shown in figure 1, line a, where the x-axis is the logarithmic value of the gas velocity, and the y-axis is the logarithmic value of the pressure drop per height of packing. [3]

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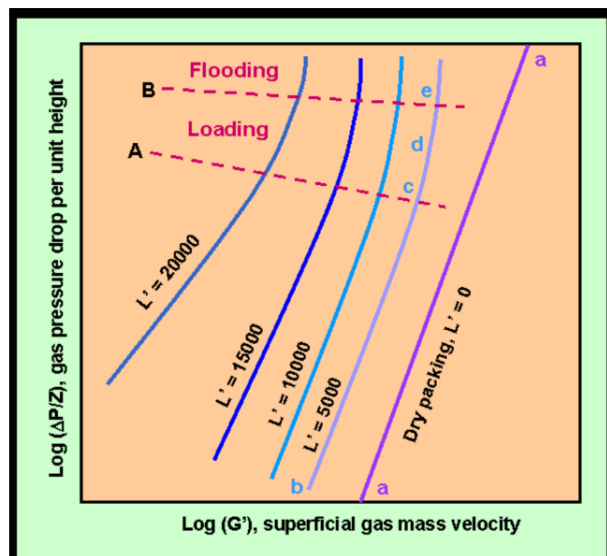


Figure 1: Fluid Characteristic in a Packed Bed Gas Absorber [3]

When the packing inside the column is wetted and a liquid flowrate is present the void volume of the packing becomes partially filled with the liquid. In turn, this reduces the cross-sectional area where gas is able to flow. Two methods for determining the void volume are suggested in this lab procedure. The easier of the two methods for determining the void volume involves measuring the amount of water necessary to fully immerse a given amount of packing. The other method requires calculation of the solid packing density and the immersed packing density (bulk density). [2] Obtaining the void volume can be done through the relationship seen in equation (1) below:

$$\epsilon = \frac{V_{void}}{V_{apparent}} \text{ or } \epsilon = 1 - \frac{\rho_B}{\rho_s} \quad (1)$$

Where:

ϵ = void volume

V_{void} = volume of the voids (cm³ or mL)

$V_{apparent}$ = total volume of the packed bed (cm³ or mL)

ρ_s = density of solid packing (g/mL)

ρ_B = bulk density of the packed particles (g/mL)

When the liquid flow is kept constant and the gas velocity is increased the relationship between the superficial gas velocity and the gas pressure drop per unit height can be observed in Figure 1, line b. At low to moderate gas velocities the behavior of the relationship is similar to that of the dry packing, which is linear. As the velocity of the gas is increased the pressure drop observed will also increase. However, at a certain point when the gas velocity continues to increase the amount of liquid retained in the packing increases significantly. This leads to a greater pressure drop and a change in slope and can be observed as point C in Figure 1, also known as the loading point. From this point, if the gas flow rate is continually increased a continuous change in the slope is also observed. Point E on the figure represents the flooding point of the column, here a sharp change in slope is observed signifying that the entire column is now filled with liquid and gas is bubbling through the packing voids. [3] The velocity of the gas observed at the flooding point is called the limiting velocity. The optimal operation of the absorption column is often observed to be at the half of the flooding velocity.[3]

In the first part of the experiment pressure drops across the packed bed gas absorber will be measured as air flow is increased until the flooding point is reached at several different liquid flowrates. This data will then be graphed in a similar fashion seen in Figure 1.

Gas Absorption Characteristics in a Packed Bed Gas Absorber

The overall mass transfer coefficient in the removal of carbon dioxide from an air stream in a packed bed gas absorber can be represented by equation (2) and (3) seen below. These equations can be applied whenever a dilute solution is being analyzed.

$$\frac{\text{Rate of mass transfer}}{\text{Volume}} = K_y a (y_A^* - y_A) \quad (2)$$

$$\frac{\text{Rate of mass transfer}}{\text{Volume}} = K_x a (x_A - x_A^*) \quad (3)$$

Where:

K_y = mass-transfer coefficient for vapor

K_x = mass-transfer coefficient for liquid

a = interface area per unit volume of the column (m^2/m^3)

*typical units for $K_y a$ and $K_x a$ are $\frac{\text{kmol}}{(\text{s} \cdot \text{m}^3)}$

x_A = bulk liquid mole fraction

y_A = bulk vapor mole fraction

y_A^* = vapor mole fraction in equilibrium with bulk liquid mole fraction (x_A)

x_A^* = liquid mole fraction in equilibrium with bulk vapor mole fraction (y_A)

The $(y_A^* - y_A)$ and $(x_A - x_A^*)$ terms are also known as the driving force and can also be partial pressure and concentration in addition to mole fractions. In this lab the concentration of CO_2 in the water samples is analyzed through titration and calculated using equation (4).

$$C_d \left(\frac{\text{gram mole}}{\text{liter of free CO}_2} \right) = \left(\frac{V_B \cdot \text{Conc. Base}}{\text{mL of Sample}} \right) \quad (4)$$

Results

Part 1. Characterization of Liquid and Gas Flow in Packed Columns

At the beginning of the lab, the void volume, ϵ , of the packed bed was determined by measuring the amount of water necessary to cover a known volume of glass solid packing. [2] A graduated cylinder was filled with the glass packing to a volume of 30 mL and then enough water was added to reach the same volume. The water was poured out into a separate graduated cylinder and measured to be 25 mL. From the ratio of these volumes, a void volume of 0.83 was determined. A summary of these values is found in Table 2 of Appendix 1 and the calculation is in Appendix III.

To find the fluid characteristics of the packed bed, the pressure drop across the column was measured at different air and water flowrates as presented in Figure 2. A summary of the flow rates and pressure drops are in Table 3 of Appendix I. Initially, the pressure drop was measured at air flow rates of 20 L/min to 70 L/min and a water flow rate of 0 L/min. The relatively constant pressure drop values indicate flooding was not achieved. There was likely an error in observing strong bubbling at the top of the column. However, at water flow rates of 1.5, 3, and 6 L/min, flooding is observed, as displayed by the red squares on Figure 2. Flood points are noted to decrease with increasing water flow. At a water flow of 1.5 L/min, the observed flood point was at an air flow of 120 L/min and decreased to 30 L/min at a water flow of 6 L/min. Between these two water flows, at 3 L/min of water, flooding occurred at 80 L/min of air flow.

In contrast to the observed flooding points, when strong bubbling of the water occurs at the top of the column, theoretical flooding occurs when there is a drastic drop in pressure. [3] This definition of flooding is displayed in Figure 2 by blue dots. At a water flow of 1.5 L/min, flooding is marked at 100 L/min of air flow, where there is a large increase in pressure drop from 7 to 13.0 cmHg. An even larger pressure drop is observed at 3 L/min of water. From 60 to 80 L/min of air, the pressure drop increases from 8.1 to 15.5 cmHg, making 60 L/min the flood point.

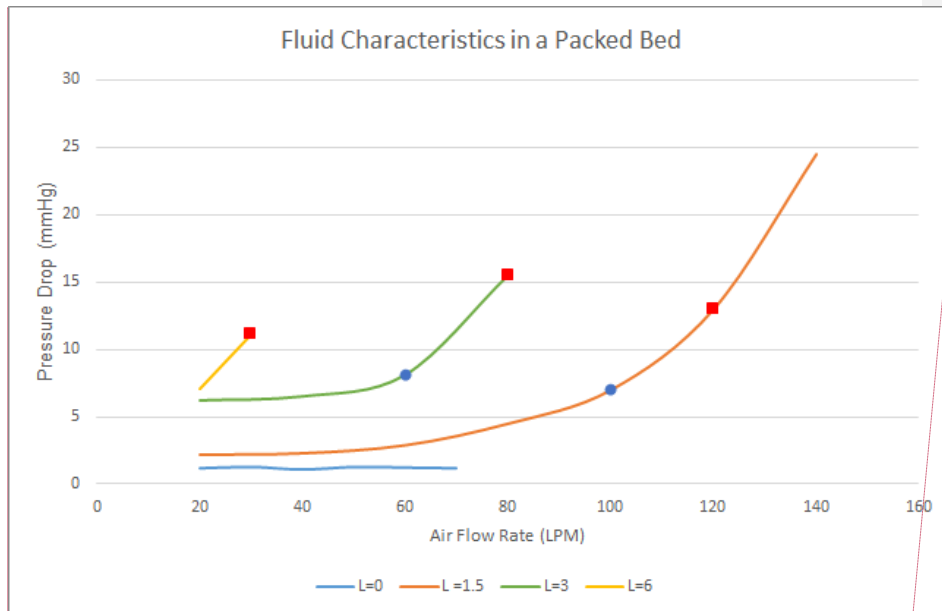
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Commented [PTH8]: By looking at your sample calculation, you used the wrong void volume. Void fraction = void volume/volume apparent. But void volume = void apparent – actual volume.

Commented [PTH9]: there is no flooding because there is no water.

Commented [PTH10]: the red squares on your figure mark the column when it is ALREADY FLOODED. not the flooding point. the flooding point is where your slope starting to increase sharply.

Commented [PTH11]: the blue dots are more likely your actual flooding point.



Commented [PTH12]: your legend of water flowrates is missing the unit. By looking at that, I wouldn't understand what L is and what the unit for that.

Figure 2. Graph of fluid characteristics showing pressure drop vs. air flow rate at varying water flow rates (L). A red square indicates observed flooding and a blue circle indicates theoretical flooding.

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Part 2a: Analysis of CO₂ in Water

This part of the experiment focuses on bringing CO₂ into the equilibrium with water. The adjusted flow rates of the air and water are 20 L/min and 6 L/min, respectively, while the flow rate of the CO₂ is kept constant at 3 L/min. To monitor the concentrations of the carbon dioxide until the equilibrium, the 50 ml volume samples are extracted and titrated in 2-minute intervals. After approximately 16 minutes the system reached equilibrium according to the concentrations measured. To visually observe behavior of the concentrations versus time the following figure is plotted:

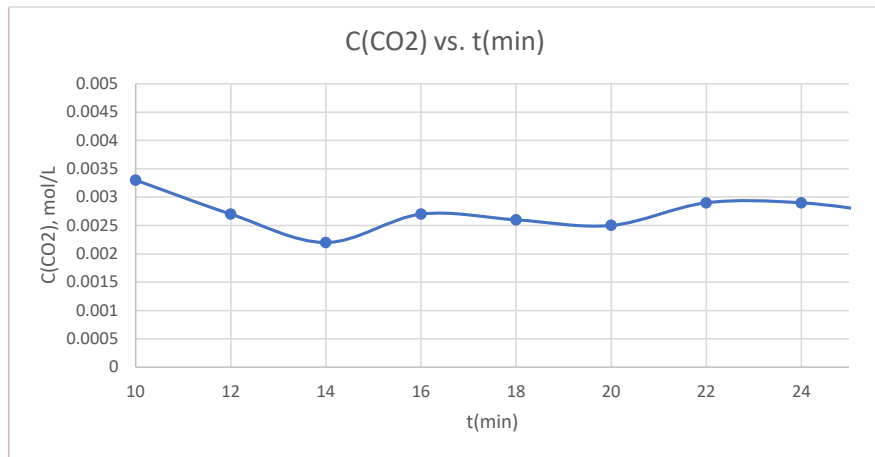


Figure 3: Curve presenting the stabilization of the carbon dioxide concentrations in water in a period of approximately 16 minutes.

Next, mass balances are used to check the accuracy of flowrate measurements and the chemical analysis. Mass IN balance result is 0.048685 g/s, while mass OUT is 0.058807 g/s, with the error percent of 20.8. Such high error points at multiple experimental errors that occurred during experiment. For the part 2a it is very important to close the system and have the same water circulate until achieved equilibrium. However, the bottom outlet water valve was open, so the water escaped the system and needed to be replenished right away. Such error could be a cause for difference between mass transfer coefficients of the bulk and the interface transport. The mass transfer coefficient times the interfacial area for absorption of carbon dioxide in water is from the gas phase to the gas-liquid interface is 0.05429. The k_a value for transport from interface to the bulk of the liquid is 2.1902×10^{-5} .

Commented [PTH14]: where is your time 0 data? that is the sample you took from the tank before CO₂ is turned on.

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Part 2b: Desorption of CO₂ in Water:

In this part of the lab, CO₂ is desorbed into air from water. Samples were then taken at 2-minute intervals until 10 samples were obtained. The samples were then titrated with 0.5N Sodium Hydroxide (NaOH) until the end point was reached.

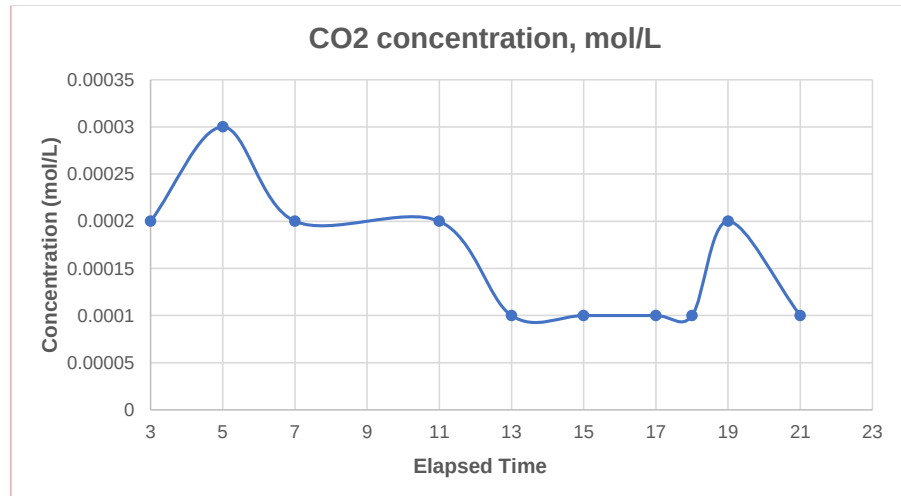


Figure 4: Desorption of CO₂ in Water

From the graph, the concentration of CO₂ in each passing sample shows an overall decreasing trend, with the exception of the sample obtained at 19 minutes which is passed off as a random error. This suggests that CO₂ was efficiently desorbed as a result. The mass transfer coefficient times the interfacial area for desorption of carbon dioxide from water is 0.00238.

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Part 3. Absorption of CO₂ by H₂O in a continuous column mode

For the absorption of CO₂ by water in a continuous column mode, the concentrations of CO₂ were determined by collecting 50 mL samples at different times intervals. Samples were collected first every 30 seconds for 3 minutes, then every minute for 5 minutes, and every 5 minutes until the concentration reached a steady state. Operating flow rates were as follows, 1.5 L/min water, 3 L/min CO₂, and 20 L/min air. After 30 minutes, a steady state was reached with the outlet concentration of CO₂ calculated as 0.0024 mol/L.

Using calculated values of the moles of CO₂ transferred, N, and the driving force, ΔC , the mass transfer coefficients times the interfacial area, k_a , was calculated to be for the gas phase to gas-liquid interface 0.0706 L/min and 0.0235 L/min for the interface to bulk of liquid. The mass transfer coefficient times the interfacial area for absorption of carbon dioxide in water in a continuous mode from the gas phase to the gas-liquid interface is 0.07060. The k_a value for transport from interface to the bulk of the liquid is 0.023553. Table 1 includes listed mass transfer coefficients for all processes tested in this experiment.

Table 1: Mass transfer coefficients calculated for all processes included in the experiment

ka interface	ka bulk	Mass transfer coefficients times the interfacial area, L/min
0.05429	2.1902E-05	Absorption of CO ₂ into water
0.00238		Desorption of CO ₂ into air
0.07060	0.023553	Absorption of CO ₂ H ₂ O in a continuous column mode

Discussion

This gas absorption experiment examined the flow characteristics in a packed bed gas absorber as well as determined the gas absorption characteristics and the overall mass transfer coefficient for removing carbon dioxide from an air stream using water flow through the packed bed absorber.

When characterizing the flow through the packed bed gas absorber the main feature studied was the flooding of the column. Flooding has been studied for absorption columns as a reference point for optimization design and occurs when there is a dramatic increase in pressure. When a column floods the efficiency of the column decreases and may damage the equipment. [5] In figure 2 below the pressure drop was graphed as a function of air flow rate at four different water flow rates. It was seen that at a water flowrate of 0 LPM the pressure drop in the column showed no dependence on the air flow rate and flooding was not achieved. This occurred because there was not water flowing through the system to restrict the flow of the air in the system. When the water flowrate was increased the pressure drop inside of the column also increased at any given air flow due to the water causing a restriction in the available area for the gas to flow through. The air flow rate at which flooding occurred was observed to decrease as the water flowrate increased. This observation aligns with the theory that when there is more restrictive air flow caused by an increase in water flow, the pressure drop dramatically increases and causes flooding of the column.

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missing discussion for part 3.

Missing comparison between part 2a and 3. this could be an interesting comparison too because you accidentally open the drain valve for part 2a. you could have mention what would happen if you close the drain valve and how your results would have change.

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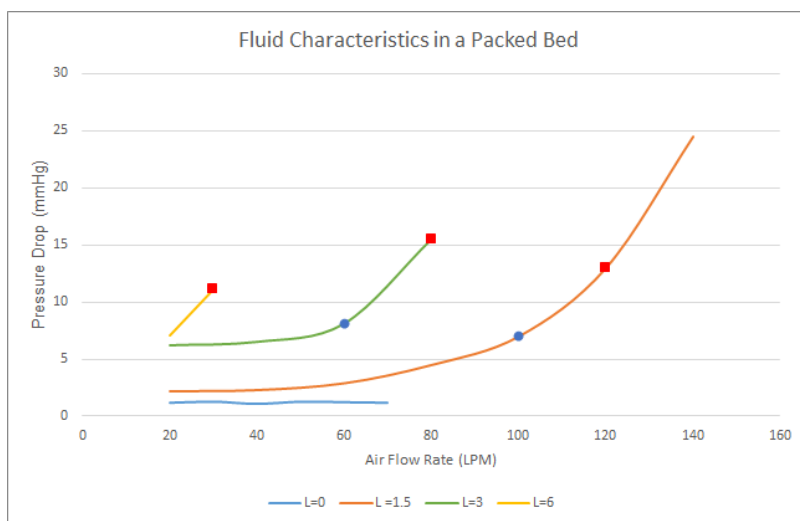


Figure 5. Graph of fluid characteristics showing pressure drop vs. air flow rate at varying water flow rates (L). A red square indicates observed flooding and a blue circle indicates theoretical flooding.

After characterizing the flow through the packed bed gas absorber, the next step was to study the gas absorption and desorption characteristics. In order to characterize the gas absorption, the carbon dioxide impurity was maintained at equilibrium with the water flowing through the column. Figure 3 below shows the concentration of carbon dioxide in the column over a duration of 24 minutes of operation. The trend shows a gradual decrease in concentration for the first couple of minutes followed by a fairly constant concentration reading with an average of 0.0027 mol/L. The unsteady nature of the plot after the 16-minute mark was probably caused by a combination of errors involving the titration end points and maintaining the liquid levels in the column explained in Appendix II. When further examined the mass flow into the column was found to have a 20% error when compared to the mass flow out of the column. This difference in flow could be from the water draw valve being open to drain while trying to maintain equilibrium. In general, the trend does seem to reach a point of constant concentration over a period of time signifying that the operation has reached steady state.

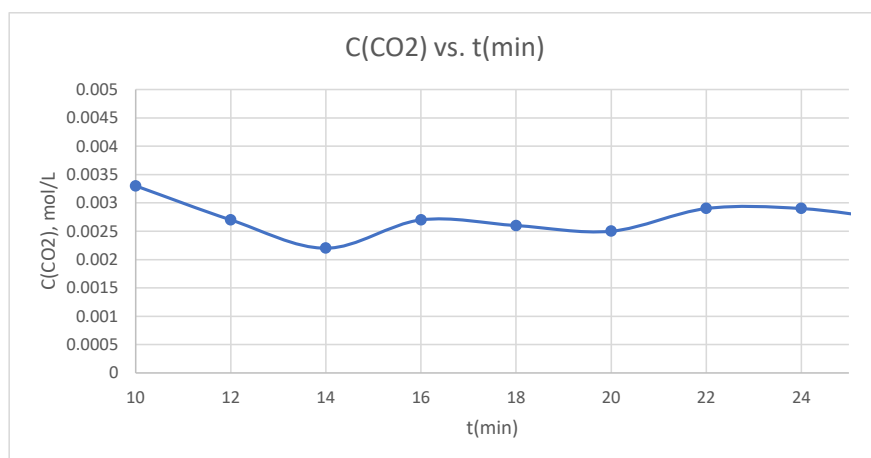


Figure 6: Curve presenting the stabilization of the carbon dioxide concentrations in water in a period of approximately 16 minutes.

Desorption of carbon dioxide from the column into the atmosphere was also studied and the trend is plotted in figure 4 below. Overall, the concentration decreases to an average of around 0.0001 mol/L showing an effective release of the free carbon dioxide into the atmosphere. The outliers may be explained through titration end point error since the concentration of carbon dioxide was much less than anticipated. A more defined trend would have come about if the samples were taken directly after each other rather than waiting 30 seconds in between sampling. The water level was also able to drop below the sampling valve a few times which may have disrupted the concentration of carbon dioxide in the sample as time was spent trying to raise the water level for sampling.

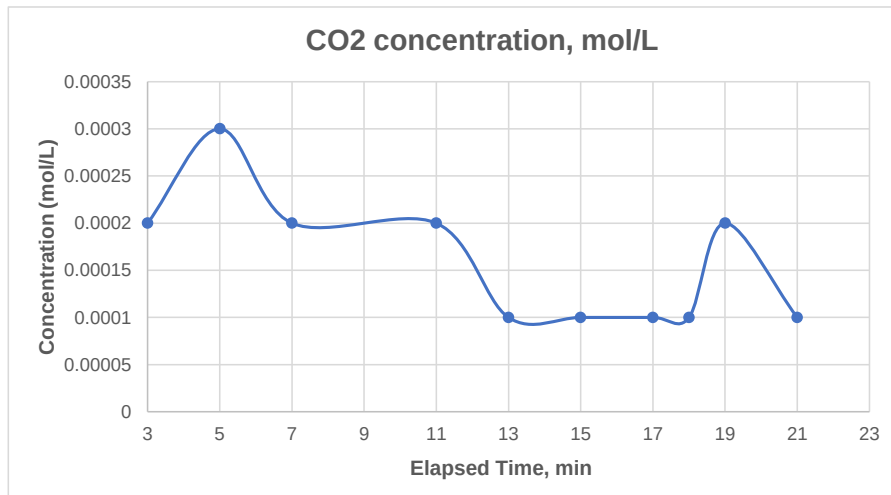


Figure 7: Desorption of CO₂ in Water

Finally, the overall mass transfer coefficients for removing carbon dioxide impurities from an air stream using water was determined to be 0.0706 L/min for the gas to gas-liquid interface and 0.0235 L/min for the interface to bulk of liquid. These were found by continuously running the column and sampling over specified time intervals. From our data it was found that the gas to gas-liquid interface transfer coefficient was dominant based on the assumption that the samples were dilute. This assumption may not be valid since with the flowing water, the interface to bulk transfer should be more effective.

Commented [PTH20]: where is the discussion for part 3?

References

Commented [PTH21]: 10/10

[1] Wankat, P.C., Chapter 13, Chapter 15, "Equilibrium Staged Separations", Prentice-Hall, Inc. 1988.

[2] Lab manual

[3] *Packed Column - Column Diameter*,
www.separationprocesses.com/Absorption/GA_Chp04a.htm.

[4] King, C.J., Chapter 11, "Separation Processes", McGraw-Hill, 1980

[5] Ajibola, Balogun K., Optimization of Flooding in an Absorption-Desorption Unit. Thesis. Central Ostrobothnia University of Applied Sciences. Degree Programme in Chemical Engineering. September 2010

Appendix I: Data Tabulation/Graphs

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1. Part I - Determination of void fraction

Table 2: Determination of the void volume of a packed bed.

Void Volume of Packed Bed		
	Solid Fill Level (mL)	Water Used (mL)
Packing	30	25
Void Fraction	0.833333333	

2. Part I - Pressure Drop Across Packed Bed

Table 3: Determination of pressure drop across a packed bed.

Pressure Drop across Packed Bed		
Air flowrate, L/min	Water flowrate, L/min	Pressure Drop (mmHg)
20	0	1.2
30		1.3
40		1.1
50		1.3
60		1.25
70		1.2
20	1.5	2.2
40		2.3
60		2.9
80		4.5
100		7.0
120		13.0
140		24.5
20	3	6.2
40		6.5
60		8.1
80		15.5
20	6	7.1
30		11.1

Figures in bold denote the point at which air flow rate the column started flooding.

3. Part II – Determination of CO₂ in Water Sample

Table 4: Summary of Part 2a operating variables

Parameter	Value	Unit
Air Flowrate	20	LPM
Water Flowrate	6	LPM
CO ₂ Flowrate	3	LPM
Temperature of Water	26.2	°C
NaOH Concentration	0.05	N
Initial CO ₂ Concentration, C ₀	0.0004	mol/L
Volume Fraction of CO ₂ in the air	0.05	-
Concentration of CO ₂ , C _{di}	0.002781	mol/L
Moles CO ₂ , N	0.000115	mol
Driving force, ΔC	8.15E-05	mol/L
Mass transfer coefficient times interfacial area (gas phase to gas-liquid interface), k _a	0.054291	LPM
Mass transfer coefficient times interfacial area (interface to bulk of liquid), k _a	2.1902E-05	LPM

Table 5. Summary of CO₂ concentrations from part 2a.

Time Lapse (min)	Conc. Base (N)	Initial Vol. of Base, (mL)	Final Vol. of Base, (mL)	Vol. of Sample (mL)	CO ₂ conc, (mol/L)
10	0.05	15	18.3	50	0.0033
12	0.05	32	34.7	50	0.0027
14	0.05	18.3	20.5	50	0.0022
16	0.05	34.7	37.4	50	0.0027
18	0.05	20.5	23.1	50	0.0026
20	0.05	37.3	39.8	50	0.0025
22	0.05	23.1	26	50	0.0029
24	0.05	39.8	42.7	50	0.0029
26	0.05	42.7	45.4	50	0.0027
Average					0.0027

Table 6: Summary of Part 2b operating variables

Parameter	Value	Unit
Air Flowrate	20	LPM
Water Flowrate	1.5	LPM
CO2 Flowrate	0	LPM
Temperature of Water	1.5	°C
NaOH Concentration	0.05	N
Initial CO2 Concentration, C_0	0.0004	mol/L
Volume Fraction of CO2 in the air	0.05	-
Concentration of CO2, C_{di}	0.00278	mol/L
Moles CO2, N	0.000134	mol
Driving force, ΔC	0.00268	mol/L
Mass transfer coefficient times interfacial area (gas phase to gas-liquid interface), k_a	0.00238	LPM
Mass transfer coefficient times interfacial area (interface to bulk of liquid), k_a	2.576E-05	LPM

Table 7: Summary of CO2 concentrations for part 2b.

Time lapse, min	Conc. Base (N)	initial Vol base, mL	Final vol base, mL	vol sample, mL	CO2 conc, mol/L
1	0.05			50	0
3	0.05	11.7	11.9	50	0.0002
5	0.05	23.3	23.6	50	0.0003
7	0.05	26.1	26.3	50	0.0002
11	0.05	23.6	23.8	50	0.0002
13	0.05	12	12.1	50	1E-04
15	0.05	23.8	23.9	50	1E-04
17	0.05	26.3	26.4	50	1E-04
18	0.05	23.9	24	50	0.0001
19	0.05	12.1	12.3	50	0.0002
21	0.05	12.3	12.4	50	1E-04

4. Part III – Determination of Absorption of CO2 by H2O

Table 8: Summary of Part 3 operating conditions.

Parameter	Value	Unit
Air Flowrate	20	LPM
Water Flowrate	1.5	LPM
CO2 Flowrate	3	LPM
Temperature of Water	26.17	°C

NaOH Concentration	0.05	N
Initial CO ₂ Concentration, C ₀	1.0x10 ⁻⁴	mol/L
Volume Fraction of CO ₂ in the air	0.04625	-
Concentration of CO ₂ , C _{di}	0.00256	mol/L
Moles CO ₂ , N	1.15x10 ⁻⁴	mol
Driving force, ΔC	1.63x10 ⁻⁴	mol/L
Mass transfer coefficient times interfacial area (gas phase to gas-liquid interface), k _a	0.0706	LPM
Mass transfer coefficient times interfacial area (interface to bulk of liquid), k _a	0.0235	LPM

Table 9: Concentration of CO₂ in water part 3.

Concentration of CO ₂ Continuous Column Data					
Sample	Time (min)	initial Vol NaOH, mL	Final vol NaOH, mL	Vb, mL	CO ₂ concentration mol/L
1	0	12.4	12.5	0.1	1E-04
2	0.5	12.5	14.8	2.3	0.0023
3	1	26.4	28.5	2.1	0.0021
4	1.5	24	26.3	2.3	0.0023
5	2	14.8	17.7	2.9	0.0029
6	2.5	28.5	31.1	2.6	0.0026
7	3	31.1	34.1	3	0.003
8	4	26.3	28.8	2.5	0.0025
9	5	17.7	19.6	1.9	0.0019
10	10	19.7	22	2.3	0.0023
11	15	22.1	24.6	2.5	0.0025
12	20	24.7	27.1	2.4	0.0024
13	25	27.2	29.8	2.6	0.0026
14	30	29.9	32.3	2.4	0.0024

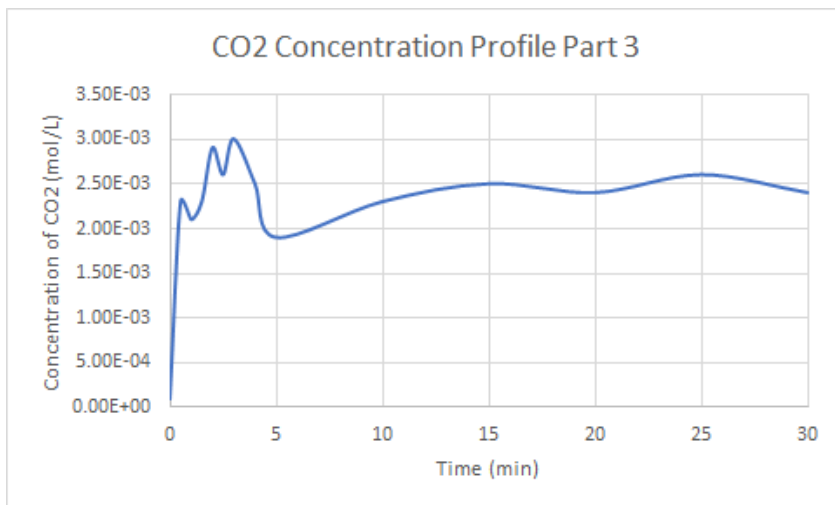


Figure 8: Concentration profile of CO₂ in water vs time for part 3.

Appendix II: Error Analysis

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Throughout the course of the experiment, there were multiple instances encountered which contributed to errors in calculations. All errors discussed in this report fall under the category of systematic errors.

1. Burette Calibration

The burettes used for titration were made out of plastic, and as such, had many bubbles that could not be extricated prior to the start of the titration process. As a result, the actual amount of Sodium Hydroxide dispensed was not the actual amount that was read off the markings on the burette.

2. Titration End Point

To indicate that the CO₂-Water mixture has been neutralized, Sodium Hydroxide was added dropwise into the conical flask until a pink color change has been observed. However, this end point was not the same across as there were 3 students who were titrating different samples at various intervals.

As such, there was a lack of consistency in endpoint determination. Furthermore, there were occasions where additional drops of Sodium Hydroxide were added in despite having reached the endpoint, resulting in more than what was required.

3. Titration Intervals

When collecting samples, the initial run necessitated the samples to be collected every 30 seconds. This meant that there was not enough time for the samples to be titrated within this short window. Hence, samples were left to sit for a period of time before titration which could have resulted in CO₂ losses. Efforts were made to curb this problem as much as possible by sealing the conical flask with parafilm.

4. Liquid level in Column

In experiments 2 and 3, the experiment manual called for the opening of the sample drain valve to retrieve samples for titration. In doing so, the fluid level had to be above the drain valve to ensure that sufficient amounts of water could be obtained. However, this proved to be difficult, as the water flow in the column was highly sensitive to the drainage valve setting.

During the experiment, it was observed that the drain valve had to be manually manipulated multiple times in an effort to maintain the water level.

5. Hempl Column Measurement

Midway through the experiment, the amount of Sodium Hydroxide in the hempl apparatus was deemed to be insufficient and had to be refilled. This attributed to inconsistent readings taken during experiment 2. Additionally, the procedure called for the operator to pull the column to the 40mL mark. However, parallax errors result in inconsistency, which creates a source of error for the actual reading.

6. Parallax Error

While attempting to take the differential pressure readings, the fluid of the red oil in the manometer was not stable; It was noticeable that the fluid level fluctuated erratically, thus causing the readings taken to be inaccurate.

7. Number of Measurements Taken

Due to the length of the experiment, there was insufficient time to take the 3 hempl column readings as outlined in the experimental manual. For this report, results reported were based off of 2 readings taken, which were found to be very similar if not the same. It was this observable trend that could lead the group to assume that readings were sufficiently consistent.

Appendix III: Sample Calculations

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1. Part I - Void Fraction Calculation

$$\epsilon = \frac{\text{Amount Of Water Added}}{\text{Volume Of Packing Added}}$$

$$\epsilon = \frac{25.0 \text{ mL}}{30 \text{ mL}} = 0.833$$

2. CO₂ Concentration in Water Sample for Parts 2(a) and 2(b)

$$C_{CO_2} = \frac{(\text{Amount of NaOH}) \times \text{Concentration of NaOH}}{\text{Volume of liquid sample}}$$

At time 10 minutes, where 3.3 mL of 0.5N Sodium Hydroxide was titrated to neutralize 50 mL of sample drawn from the column, the concentration of CO₂ in the sample could be calculated:

$$C_{CO_2} = \frac{(3.3 \text{ mL}) \times 0.5N}{50 \text{ mL}} = 0.0033 \text{ mol/L}$$

3. Amount of free CO₂ water

v – volume fraction for 3 L/min CO₂

$$v = \frac{V_{\text{measured}}}{V_{\text{piston}}} = \frac{2 \text{ ml}}{40 \text{ ml}} = 0.05$$

4. Inlet concentration of CO₂

$$PV = nRT, C = \frac{n}{V}, C = \frac{vP}{RT}$$
$$C_{di} = \frac{0.05 * 137.8952 \text{ kPa}}{8.314 \frac{\text{LkPa}}{\text{molK}} * 298.15 \text{ L}} = 0.002781 \text{ mol/L}$$

5. Mass transfer coefficient calculations

$$N = ka\Delta C$$

N – moles of CO₂ transferred

ka – mass transfer coefficient * interfacial area

ΔC – driving force

$$\Delta C = C_0 - C_{di} = 0.0027 - 0.002781 \frac{\text{mol}}{\text{L}} = 8.15E - 05 \frac{\text{mol}}{\text{L}}$$

$$N = (C_{d0} - C_0)V = \frac{(0.0027 - 0.0004)mol}{L} * 0.05 = 0.000115mol$$

$$t = 26 \text{ min}$$

$$ka = \frac{N}{t\Delta C} = \mathbf{0.054291L/min}$$

7. Mass flowrate of CO₂ at inlet

$$m_{out} = \frac{PF_{out}v M_w}{RT} = \frac{137.8952 * \left(\frac{23}{60}\right) * 0.05 * 44.01}{8.3134 * 298.15} = \mathbf{0.046925 \frac{g}{s}}$$

$$m_{in} = \mathbf{0.04685 \frac{g}{s}}$$

$$mass \text{ in} = mass \text{ out}$$

$$mass \text{ out} = m_{out} + F_{CO_2}C_{d0}M_w = \mathbf{0.058807 \frac{g}{s}}$$

$$mass \text{ in} = m_{in} + F_{water}C_0M_w = \mathbf{0.048685 \frac{g}{s}}$$

The desorption and the mass coefficients from the interface to the bulk of the liquid is calculated using the same equations and calculations as above.

Appendix IV: Individual Team Contributions

NAME: GUZIK, ALICE

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	2	
Experimental Objective (Prelab)	-	
Apparatus (prelab)	1.5	
Materials & Supplies (prelab)	-	
Procedure (prelab)	-	
Project Safety Evaluation (prelab)	-	
Final Lab Editing	1.5	
Executive Summary (final lab)	-	
Introduction (Final lab)	1	
Theory (Final Lab)	2.5	
Results (final lab)	-	
Discussion (final lab)	-	
Conclusion (final lab)	-	
References (final and/or prelab)	0.5	
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)	-	
Sample Calculations (final Lab)	-	
Power Point Presentation	1.5	
Total	16.5	

NAME: **HADZIC, ADISA**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	2	
Experimental Objective (Prelab)	2	
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	5	
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)		
Sample Calculations (final Lab)	3	
Power Point Presentation	2	
Total	21	

NAME: **KRUPA, KELLEN**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	2	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	2	
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	2	
Executive Summary (final lab)	1	
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	6	
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	3	
Error Analysis (final lab)		
Sample Calculations (final Lab)	3	
Power Point Presentation	1	
Total	26	

NAME: **LIM, IAN**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	Titration
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	-	
Materials & Supplies (prelab)	-	
Procedure (prelab)	-	
Project Safety Evaluation (prelab)	1.5	
Final Lab Editing	1.5	
Executive Summary (final lab)	-	
Introduction (Final lab)	-	
Theory (Final Lab)	-	
Results (final lab)	1	
Discussion (final lab)	-	
Conclusion (final lab)	-	
References (final and/or prelab)	0.5	
Data Tabulation/Graphs (final)	-	
Error Analysis (final lab)	2	
Sample Calculations (final Lab)	3	
Power Point Presentation	1.5	
Total	18	

NAME: **MIRANDA, ALEXIS**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)		
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		
Total		